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Journal 1

Journal Title: The Effectiveness of Capture the Flag as a Network Security Practical Intervention on Students' Self-Efficacy and Learning Outcomes

Author(s): A. St. Khadijah Ridwan, Ummu Radiyah, Muhammad Ihksan Malik, Muhammad Nauval Pattiroi, Muslika, Nining Ulul Fajri, and Nirwana Amir

Link to the Journal: <https://journal.unm.ac.id/index.php/INTEC/article/view/11523>

APA Citation: Ridwan, A. S. K., Radiyah, U., Malik, M. I., Pattiroi, M. N., Muslika, Fajri, N. U., & Amir, N. (2025). *The effectiveness of Capture the Flag as a network security practical intervention on students' self-efficacy and learning outcomes. Information Technology Education Journal*, 4(4), 811-824.
<https://doi.org/10.59562/intec.v4i4.11191>

No	Components	Description
1	Research question/problem statement	This study asks whether participation in a Capture the Flag (CTF) competition can improve students' self-efficacy and learning outcomes in a network security course when compared with traditional practical exercises. The article addresses the problem that cybersecurity teaching may build technical knowledge but not always strengthen confidence or clearly measure the impact of gamified practice.
2	Motivation/relevance	The authors argue that cybersecurity education needs practical and engaging strategies that prepare students for real-world challenges. Because CTF activities are already popular in training environments but their psychological effects are less discussed, the study is relevant for lecturers who want evidence before integrating gamified tasks into the curriculum.
3	Theoretical framework	The article is grounded mainly in Bandura's self-efficacy theory and in ideas from gamification and active learning. It assumes that mastery experiences, immediate feedback, competition, and authentic problem solving can increase students' confidence while also improving their understanding of network security concepts.
4	Method	The researchers use a quasi-experimental pre-test/post-test control-group design with 60 undergraduate students in a network security course. The experimental group joins a two-week CTF competition, while the control group completes instructor-guided practical exercises; data come from a self-efficacy scale, a final exam, and a practical skills assessment.
5	Results/arguments	The CTF group improves more strongly than the control group on all major measures. Their self-efficacy rises from 3.2 to 4.0, final exam scores reach 85% compared with 75% in the control group, practical skills scores reach 88% compared with 78%, and engagement is also reported as higher.
6	Conclusions	The study concludes that CTF competitions are an effective practical intervention for increasing student confidence, conceptual understanding, hands-on skills, and engagement in network security education. At the same time, the authors acknowledge that the small sample and single-institution setting limit how far the findings can be generalized.
7	Evaluation	This is a useful and recent academic source because it presents clear hypotheses, measurable outcomes, and direct comparison with a control group. However, the quasi-experimental design, short intervention period, and relatively small sample reduce the strength of its causal claims and limit wider generalization.
8	Analysis	This source will be very helpful for supporting an argument that CTF-based learning can improve both performance and self-confidence. I can use it as one of the key empirical studies in the bibliography, especially when discussing how cybersecurity activities affect self-efficacy, engagement, and practical achievement.

Journal 2

Journal Title: Analysis of Cyber Security Knowledge and Skills for Capture the Flag Competition

Author(s): Jean Tirstan T, Joy Gilbert A, and Nelmiawati

Link to the Journal: <https://jurnal.polibatam.ac.id/index.php/JI/article/view/3986>

APA Citation: Tirstan, J., Gilbert, J., & Nelmiawati. (2022). Analysis of cyber security knowledge and skills for Capture the Flag competition. *Jurnal Integrasi*, 14(1), 14-22. <https://doi.org/10.30871/ji.v14i1.3986>

No	Components	Description
1	Research question/problem statement	This study investigates what knowledge and technical skills are most needed in CTF competitions and how those findings can be used to design a better learning platform for cybersecurity students. The article addresses the problem that beginners often struggle with platform complexity and do not always receive structured skill development.
2	Motivation/relevance	The authors are motivated by the rapid growth of academic CTF competitions and the need to ensure that these competitions genuinely support cybersecurity education. The topic is relevant because it connects CTF design, beginner support, and curriculum planning for student training groups such as the Polibatam Cyber Team.
3	Theoretical framework	The paper treats CTF as a form of challenge-based, skill-based, and technique-based learning. It organizes cybersecurity education around major CTF categories such as coding, cryptography, forensics, web, reverse, and pwn, while also emphasizing progressive difficulty levels of easy, medium, and hard.
4	Method	The researchers collect and analyze about 500 CTF solutions and write-ups from CTFtime and GitHub, then map the skills and techniques used in each challenge category. They use this analysis to build a CTF platform for PCT regeneration and observe participant progress, performance, and feedback.
5	Results/arguments	The study argues that successful CTF training depends on category-specific skill practice, easier platform setup, and gradual difficulty progression. It reports that the skill- and technique-based platform helps around 80% of participants improve, with 40 of 50 students showing progress within one month.
6	Conclusions	The article concludes that CTF competitions can significantly support cybersecurity education when they are designed around explicit skills, techniques, and progressive challenge levels. It also suggests that competition is an effective way to motivate learners and build practical competence over time.
7	Evaluation	This source is useful because it combines a large collection of CTF write-ups with a practical educational application in a student training platform. However, some parts of the method are described quite generally, and the paper relies more on descriptive analysis and reported improvement than on rigorous experimental comparison.
8	Analysis	I can use this source to explain what kinds of knowledge and skills CTF activities develop and why challenge design matters in cybersecurity education. It complements the first journal well because it provides background on skill categories and platform design, while the first journal offers stronger evidence about learning outcomes and self-efficacy.